

Vol 8 Chapter 4 — Hamiltonian Mechanics

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 4 — Hamiltonian Mechanics

Level 1 — one sentence

Hamiltonian mechanics (Hamilton 1834) is the phase-space formulation of classical mechanics: replace velocities $\dot{q}$ with canonical momenta $p = \partial L / \partial \dot{q}$, define the Hamiltonian $H(q, p, t) = \sum p_i \dot{q}_i - L$, and dynamics is governed by Hamilton's first-order equations $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$ — the foundation for canonical quantization in QM (Vol 5).

Level 2 — graduate-physicist precision

4.1 From Lagrangian to Hamiltonian: Legendre transform

Define canonical momentum: $p_i = \partial L / \partial \dot{q}_i$.

Hamiltonian: $H(q, p, t) = \sum_i p_i \dot{q}_i - L(q, \dot{q}, t)$ where $\dot{q}_i$ is solved in terms of p_i .

For $L = (1/2)m\dot{q}^2 - V(q)$: $p = m\dot{q}$, $\dot{q} = p/m$, $H = p^2/(2m) + V(q) = T + V = \text{total energy}$.

4.2 Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

These are $2n$ first-order ODEs (instead of n second-order from Lagrangian).

4.3 Phase space and Liouville's theorem

The system's state is a point in phase space $(q_1, \dots, q_n, p_1, \dots, p_n)$ of dimension $2n$. Time evolution carries each initial point along a unique trajectory.

Liouville's theorem: the phase-space volume is preserved by Hamiltonian flow. $d/dt[\text{vol}] = 0$ along trajectories. Has deep statistical-mechanics consequences (foundation for ergodic hypothesis, Vol 6 Ch 6).

4.4 Poisson brackets

For two phase-space functions $f(q, p)$, $g(q, p)$:

$$\{f, g\}_{PB} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

Properties: antisymmetric, Jacobi identity, Leibniz rule. Hamilton's equations:

$$\dot{f} = \{f, H\} + \partial_t f$$

If $\{f, H\} = 0$ and f has no explicit time dependence, f is conserved.

Fundamental: $\{q_i, p_j\} = \delta_{ij}$.

4.5 Bohr correspondence

Replacement $\{f, g\}_{PB} \leftrightarrow [\hat{f}, \hat{g}]/(i\hbar)$ — classical Poisson bracket becomes quantum commutator divided by $i\hbar$. Bohr correspondence principle. The canonical commutator $[\hat{q}_i, \hat{p}_j] = i\hbar\delta_{ij}$ (Vol 5 Ch 2) is the quantum version of the classical $\{q_i, p_j\} = \delta_{ij}$.

Substrate reading: the Poisson bracket structure is the substrate's classical-limit Lie algebra of phase-space functions; the quantum commutator structure is the substrate's full operator algebra. They map to each other at Scale 1 $\leftrightarrow$ Scale 2 transition.

4.6 Canonical transformations

A transformation $(q, p) \rightarrow (Q, P)$ is canonical if it preserves the symplectic structure (Poisson brackets). Useful for solving problems by choosing coordinates that make the Hamiltonian simpler.

Generating functions (4 types: $F_1(q, Q), F_2(q, P), F_3(p, Q), F_4(p, P)$) give explicit canonical transformations.

4.7 Hamilton-Jacobi equation

For action $S(q, t)$:

$$H(q, \partial S / \partial q, t) + \partial S / \partial t = 0$$

Solving gives the action; classical trajectories are characteristics. Hamilton-Jacobi is the closest classical analog of the Schrödinger equation; the substitution $\psi \sim e^{iS/\hbar}$ converts H-J to Schrödinger in the WKB limit.

4.8 K-audit anchors

- Vol 5 Ch 2: position/momentum commutator from substrate
- Vol 5 Ch 4: Schrödinger equation from substrate Casimir
- Vol 6 Ch 6: Liouville's theorem foundation for ergodic hypothesis

Level 3 — 5th-grader accessibility

Hamiltonian mechanics is mechanics rewritten in phase space — using positions q AND momenta p as independent variables (instead of q and $\dot{q}$). Hamilton's equations are first-order: $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q$. The Hamiltonian $H = T + V$ is usually total energy. Liouville's theorem: phase-space volume is preserved by Hamilton's flow. Poisson brackets generalize to quantum commutators via Bohr correspondence (this is canonical quantization). The Hamiltonian formalism is the natural language for both classical statistical mechanics and quantum mechanics — that's why physicists prefer it for advanced topics.

What comes next

Chapter 5 develops symmetries and Noether's theorem — the substrate's parent symmetry framework.

Where to look this up

- Goldstein Ch 8; Arnold, *Mathematical Methods of Classical Mechanics*
- BST: Vol 5 Ch 2 + Ch 4; Vol 6 Ch 6